



Pandas

Pandas is a powerful data analysis and manipulation library for Python. This cheat sheet is a quick reference for Pandas beginners.



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Getting Started

Introduction

You'll need to import pandas to get started:

```
import pandas as pd
```

Creating DataFrames

```
pd.DataFrame(data={'col1': [1, 2],  
                  'col2': [3, 4]})
```

From a dictionary

```
pd.DataFrame(data=[{'a': 1, 'b':  
2}, {'a': 3, 'b': 4}])
```

From a list of dictionaries

```
pd.read_csv('file.csv')
```

From a CSV file

```
pd.read_excel('file.xlsx')
```

From an Excel file

Inspecting Data

```
df.head()
```

First 5 rows

```
df.tail()
```

Last 5 rows

```
df.shape
```

Number of rows and columns

```
df.info()
```

Info on DataFrame

```
df.describe()
```

Summary statistics

```
df.columns
```

Column names

```
df.index
```

Index

```
df.dtypes
```

Data types of columns

Selecting Data

```
df['col1']
```

Select column

```
df[['col1', 'col2']]
```

Select multiple columns

```
df.loc[0]
```

Select row by index

```
df.loc[:, 'col1']
```

Select all rows for 'col1'

```
df.iloc[0]
```

Select row by position

```
df.iloc[0, 1]
```

Select specific value

```
df[df['col1'] > 2]
```

Select rows based on condition

Data Cleaning

```
df.dropna()
```

Drop rows with any missing values

<code>df.dropna(axis=1)</code>	Drop columns with any missing values
<code>df.fillna(0)</code>	Replace missing values with 0
<code>df.drop_duplicates()</code>	Drop duplicate rows
<code>df.rename(columns={'old_name': 'new_name'})</code>	Rename columns

Adding/Removing Data

<code>df['col3'] = df['col1'] + df['col2']</code>	Add new column
<code>df.drop('col1', axis=1)</code>	Drop column
<code>df.append(new_row)</code>	Add new row
<code>df.insert(2, 'new_col', new_data)</code>	Insert new column at position 2

Combining Data

<code>pd.concat([df1, df2])</code>	Concatenate rows
<code>pd.concat([df1, df2], axis=1)</code>	Concatenate columns
<code>pd.merge(df1, df2, on='key')</code>	Merge DataFrames on key
<code>pd.merge(df1, df2, left_on='key1', right_on='key2')</code>	Merge on different keys
<code>df1.join(df2, lsuffix='_left', rsuffix='_right')</code>	Join DataFrames

Aggregating Data

<code>df['col1'].sum()</code>	Sum of values in column
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<code>df['col1'].mean()</code>	Mean of values in column
<code>df['col1'].count()</code>	Count of values in column
<code>df['col1'].min()</code>	Minimum value in column
<code>df['col1'].max()</code>	Maximum value in column
<code>df['col1'].std()</code>	Standard deviation
<code>df['col1'].var()</code>	Variance
<code>df.groupby('col1').sum()</code>	Group by and sum
<code>df.groupby('col1').mean()</code>	Group by and mean
<code>df.groupby(['col1',</code>	Group by multiple columns

Applying Functions

<code>df.apply(np.sqrt)</code>	Apply function to all values
<code>df['col1'].apply(lambda x: x ** 2)</code>	Apply function to column
<code>df.applymap(str)</code>	Apply function to DataFrame elements
<code>df['col1'].map({'a': 1, 'b': 2})</code>	Map values
<code>df['col1'].replace('a', 1)</code>	Replace values

Handling Dates

<code>df['date'] = pd.to_datetime(df['date'])</code>	Convert to datetime
<code>df['year'] = df['date'].dt.year</code>	Extract year
<code>df['month'] = df['date'].dt.month</code>	Extract month
<code>df['day'] = df['date'].dt.day</code>	Extract day

```
df.set_index('date', inplace=True)
```

Set date as index

Input/Output

```
df.to_csv('file.csv')
```

Save DataFrame to CSV

```
df = pd.read_csv('file.csv')
```

Load DataFrame from CSV

```
df.to_excel('file.xlsx')
```

Save DataFrame to Excel

```
df = pd.read_excel('file.xlsx')
```

Load DataFrame from Excel

```
from sqlalchemy import  
create_engine
```

Import SQLAlchemy for SQL operations

```
engine =  
create_engine('sqlite:///memory:'  
)
```

Create SQL engine

```
df.to_sql('table_name', engine)
```

Save to SQL table

```
df = pd.read_sql('table_name',  
engine)
```

Load from SQL table

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